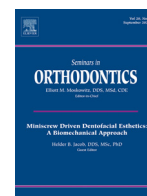




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# Integrative omics analyses identified a damaging protein–altering variant in *KIF1B* contributing to the risk of craniofacial microsomia

Waheed Awotoye<sup>a,b,c</sup>, Pedro A. Sanchez–Lara<sup>d</sup>, Stephen Yen<sup>b,e</sup>, Emily Farrow<sup>f</sup>, Huojun Cao<sup>a</sup>, Aline Petrin<sup>a</sup>, Lina Moreno Uribe<sup>a,c</sup>, Veerasathpurush Allareddy<sup>g</sup>, Shankar Rengasamy Venugopalan<sup>c,h,\*</sup>

<sup>a</sup> Iowa Institute for Oral Health Research, University of Iowa, Iowa City, IA, USA

<sup>b</sup> Children's Hospital Los Angeles, Los Angeles, CA, United States

<sup>c</sup> Department of Orthodontics, University of Iowa College of Dentistry and Dental Clinics, Iowa City, IA, USA

<sup>d</sup> Department of Pediatrics, Guerín Children's at Cedars–Sinai Medical Center, Los Angeles, CA, USA

<sup>e</sup> Center for Craniofacial Molecular Biology, University of Southern California, Los Angeles, CA, USA

<sup>f</sup> Department of Pathology, Children's Mercy Hospital, University of Missouri–Kansas City School of Medicine, Kansas City, MO, USA

<sup>g</sup> Department of Orthodontics, UIC College of Dentistry, Chicago, IL, USA

<sup>h</sup> Department of Orthodontics, Tufts University School of Dental Medicine, Boston, MA, USA

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## ABSTRACT

**Background:** Craniofacial microsomia (CFM) is the second most common facial birth defect characterized by anomalies in structures that develop from the first and second pharyngeal arches. Most cases are spontaneous with unknown etiology, and the overall contribution of genetic risk factors remain to be explored. This study used integrative omics to identify pathogenic variants in the exome sequenced dataset of probands and families with CFM.

**Method:** The exome sequence dataset used in this study was generated from buccal swab DNA of probands diagnosed with CFM. The dataset also included sequenced data from mother or father who were unaffected. We used integrative omics approach consisting of epigenomics through human craniofacial super–enhancers (hCFSEs) enrichment analysis and exome sequencing analysis to identify pathogenic genetic variants that would explain the developmental pathogenesis of this syndrome in families.

**Results:** Our epigenomic analysis found significant enrichment of CFM candidate genes within the hCFSEs. Through integrative ‘omics’ approach (epigenomics and exome sequencing), we identified a damaging protein–altering variant in *KIF1B* (c.2525T>C). Mouse transcriptomic analysis showed that *Kif1b* is expressed in pharyngeal arches and knockout mice had craniofacial anomalies including ear defects. Further bioinformatic analyses showed that the variant which lies in a critical functional domain, disrupts KIF1B interactome and likely disrupting protein function.

**Conclusion:** Through an integrative omics approach we identified a likely pathogenic variant in a CFM candidate gene, *KIF1B*. This robust approach has the potential to identify new CFM etiologic variants and further our knowledge of the molecular diagnosis of this developmental anomaly.

## Introduction

Craniofacial microsomia (CFM) is the second most common facial congenital anomaly, with an estimated incidence of 1 in 3500 to 1 in 5600 live births.<sup>1</sup> It results from abnormal development of the first and second pharyngeal arches, leading to malformations of derivative craniofacial structures. Clinical manifestations of CFM include unilateral hypoplasia of the mandible and ear (e.g., shortened mandibular ramus, small glenoid fossa, malformed condyle, preauricular tags, microtia, and cupped ear). In severe cases, hearing loss may occur due to atresia or stenosis of the external auditory canal. Additional features may include

orbital malformations (such as microphthalmia and anophthalmia), facial palsy, cleft lip and/or palate, and soft tissue deficiencies affecting the masticatory muscles and parotid gland.

Given the broad range of craniofacial structures affected and the risk of failure to thrive in severe cases with mandibular hypoplasia, CFM represents a significant public health concern. Management primarily involves surgical correction of structural anomalies, frequently including reconstruction of the ears, jaws, oral cavity, and temporomandibular joint (TMJ).<sup>2</sup> Environmental risk factors implicated in CFM pathogenesis include maternal smoking, gestational diabetes, use of vasoactive medications, thalidomide, and retinoic acid. Genetic factors also play a

\* Corresponding author at: Professor of Dental Medicine, Tufts School of Dental Medicine, One Kneeland Street, Boston, MA 02111, USA.

E-mail addresses: awotoye@uiowa.edu (W. Awotoye), Shankar.RV@Tufts.edu (S. Rengasamy Venugopalan).

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substantial role.<sup>3–5</sup> Evidence from sporadic cases supports the contribution of germline *de novo* mutations, while familial cases indicate autosomal dominant inheritance with incomplete penetrance and variable expressivity.<sup>3,6–10</sup> Genetic association studies, linkage analyses, and candidate gene approaches have identified putative risk genes.<sup>11–15</sup> Recent next-generation sequencing studies have implicated genes such as *MYT1*, *SF3B2*, *ITPR1*, and *AMIGO2* in CFM etiology.<sup>10,16–18</sup>

Super-enhancers are highly active and cell type-specific regulatory elements and are critical determinants of craniofacial morphogenesis through their roles in cell fate specification and gene expression regulation.<sup>19</sup> Among these, craniofacial super-enhancers have been shown to direct facial morphogenesis, and disruptions in these elements can lead to craniofacial anomalies.<sup>20,21</sup> Pathogenic variants in genes regulated by these super-enhancers have also been implicated in common craniofacial birth defects.<sup>22</sup> Therefore, systematic investigation of pathogenic variants in genes enriched in craniofacial super-enhancers may facilitate the identification of new CFM candidate genes. In this study, we employed an integrative approach combining an exome-sequenced dataset of CFM families with enrichment for genes harboring protein-altering variants in craniofacial super-enhancers, thereby identifying new risk genes that may contribute to the developmental pathogenesis of CFM.

## Materials and methods

### Enrichment of CFM genes in human craniofacial super-enhancers

Transcriptional regulation of genes essential for craniofacial morphogenesis is mediated by human craniofacial super-enhancers (hCFSEs).<sup>22</sup> To assess whether CFM-associated genes are enriched among those regulated by hCFSEs, we performed an enrichment analysis using the publicly available hCFSE dataset (NCBI GEO GSE97752).<sup>22</sup> Statistical significance was assessed, and results were visualized using a scatterplot with  $-\log_{10}(p\text{-value})$  on the x-axis and odds ratio on the y-axis. A significance threshold of  $p < 0.05$  was employed.

To curate a high-confidence list of human CFM candidate genes, we systematically queried the human gene database (<https://www.geneCards.org/>) using the keywords “craniofacial microsomia,” “hemifacial microsomia,” and “microsomia.” We then cross-referenced these results with gene-disease associations from multiple databases, including ClinVar (<https://www.ncbi.nlm.nih.gov/clinvar/>), DISEASES (<https://diseases.jensenlab.org/>), OMIM® (<https://omim.org/>), UniProtKB/Swiss-Prot (<https://www.uniprot.org/>), the GWAS Catalog (<https://www.ebi.ac.uk/gwas/>), GeneHancer, and the Human Phenotype Ontology (<https://hpo.jax.org/app/>).<sup>23–25</sup> Additionally, we performed a comprehensive review of published peer-reviewed literature to identify genes potentially absent from the integrated GeneCards database. Recognizing the high sensitivity but limited specificity of the human gene database, we refined our candidate list by retaining only those genes with robust evidence of association with CFM, specifically those supported by human genetic studies (e.g., GWAS, next-generation sequencing with functional validation) as documented in the literature.

Secondarily, we included this analysis to identify new genes that play roles in craniofacial development whose disease-causing variants have not been reported in CFM pathogenesis. These genes are regulated by the hCFSEs thus, their spatiotemporal specificities strongly suggest their role in craniofacial development.<sup>22</sup>

### Analysis of CFM exome sequencing data

The exome sequence dataset used in these analyses were generated from buccal swab DNA of CFM probands (7 right-sided, 2 left-sided, 1 bilateral), with diagnoses confirmed by a board-certified pediatric medical geneticist. Unaffected parental DNA, including maternal samples for all probands and a paternal sample for one proband, was also part of the sequenced data. The study protocol, encompassing exome data generation and primary bioinformatic analysis, was approved by the

Institutional Review Boards of Children’s Mercy Hospital and Clinics (#15050169) and the University of Missouri Kansas City (Request-to-Rely). Secondary bioinformatic analyses were approved by the University of Iowa Institutional Review Board (#201905851).

Genomic DNA was isolated, and exonic regions were captured and enriched using the Integrated DNA Technologies (IDT) panel. Exome sequencing was performed on the Illumina HiSeq 2500 platform using TruSeq v4 reagents, generating paired-end 125-nucleotide reads. The mean sequencing coverage was 84X (Fig. 1).

Sequencing reads were aligned to the human reference genome (GRCh38/hg38) using the Burrows–Wheeler Aligner (BWA), generating binary alignment map (BAM) files (Fig. 1).<sup>26</sup> Variant calling was performed with the Genome Analysis Toolkit (GATK) workflow, including HaplotypeCaller in GVCF mode and GenotypeGVCFs, to identify single nucleotide polymorphisms (SNPs) and short insertions and deletions (InDels).<sup>27</sup> Variants were annotated using the RefSeq database based on the hg38 reference genome (<https://www.ncbi.nlm.nih.gov/projects/genome/guide/human/index.shtml>) (Fig. 1).

### Analysis of high confidence rare protein-altering coding variants

To identify pathogenic variants contributing to the developmental pathogenesis of CFM in these families, we screened the entire coding region for rare protein-altering variants (Fig. 1). Variants were first filtered for high confidence, defined as genotype quality  $\geq 20$  and read depth  $\geq 10$ .

Among high-confidence variants, we prioritized those with a minor allele frequency (MAF)  $< 1\%$  as reported in the Genome Aggregation Database (gnomAD; <https://gnomad.broadinstitute.org/>).<sup>28</sup> Rare variants were further assessed for their predicted impact on protein function. Variants causing nonsynonymous amino acid substitutions (missense), frameshifts due to insertions or deletions (InDels), or loss-of-function (LOF) effects were prioritized for downstream analysis.<sup>28</sup>

### Investigation of damaging protein-altering variants

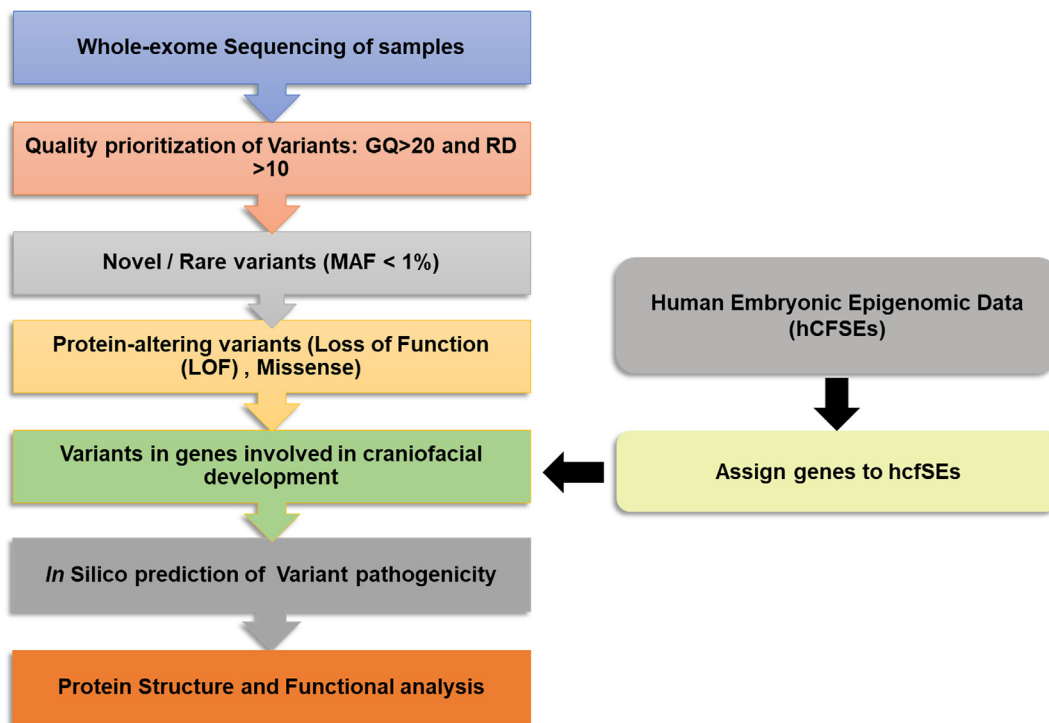
Many rare variants within the coding region of the genome do not result in birth defects.<sup>29</sup> To identify disease-causing variants underlying craniofacial defects in these individuals, we assessed the pathogenicity of protein-altering variants using bioinformatic prediction tools. These in silico tools prioritize variants based on their predicted deleterious effects on protein structure and function (Fig. 1).<sup>30–32</sup>

We primarily used three in silico prediction tools to assess variant pathogenicity: Sorting Intolerant From Tolerant (SIFT; <http://sift.jcvi.org/>), Polymorphism Phenotyping v2 (PolyPhen2; <http://genetics.bwh.harvard.edu/pph2/>), and Combined Annotation Dependent Depletion (CADD; <https://cadd.gs.washington.edu/>). Clinical significance of variants was further evaluated using ClinVar (<https://www.ncbi.nlm.nih.gov/clinvar/docs/clinvar/>), which classifies variants as benign, of uncertain significance, or pathogenic based on multiple lines of evidence.<sup>33–35</sup> These tools are integrated within the Varsome platform (<https://varsome.com/>), which we used as a comprehensive resource for variant annotation and classification.<sup>36</sup>

To assess the structural and functional impact of pathogenic variants, we used the Help yOur Protein Explained (HOPE) tool (<https://www3.cmbi.umcn.nl/hope/>) (Fig. 1),<sup>37</sup> which analyzes the consequences of amino acid substitutions on protein structure. Wild-type protein structures were modeled using AlphaFold2 (<https://alphafold.ebi.ac.uk/>) (Fig. 1),<sup>38</sup> and mutant protein sequences were generated to incorporate the identified amino acid changes. We then evaluated the effects of these substitutions on protein structure and physicochemical properties.

### Prioritization of protein-altering variants in genes enriched in hCFSEs

Here, we prioritized those pathogenic variants that are found in genes with evidence of roles in craniofacial development.



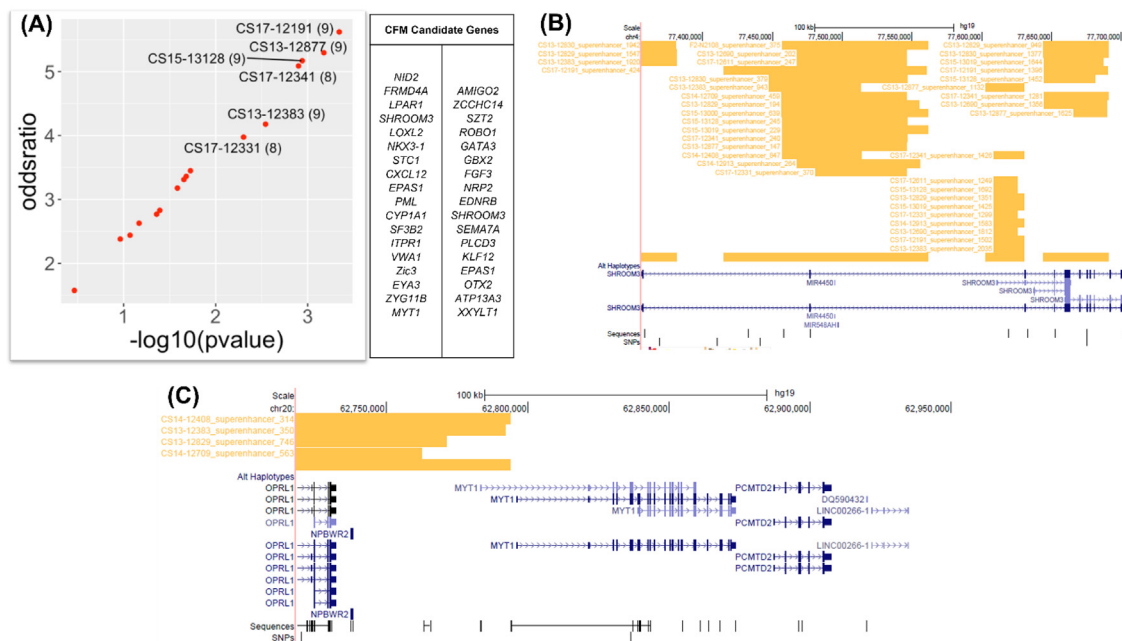
**Fig. 1.** Integrative OMICS work-flow of analyses for identification pathogenic variants contributing to CFM risk in our cohort; \*GQ: Genotype Quality; \*RD: Read Depth; \*MAF: Minor Allele Frequency; \*hcfSEs: human craniofacial super-enhancers.

These genes were selected based on regulation by hCFSEs (Fig. 1). Also, we investigated the phenotype resulting from the knockout in mice using the Mouse Genome Informatics, MGI database (<https://www.informatics.jax.org/>) as well as investigated the expression of these genes within the pharyngeal arches in mice.<sup>39</sup>

## Results

### CFM candidate genes enriched within the putative targets hCFSEs

Firstly, we investigated the enrichment of known CFM genes within the putative targets of hCFSEs (Fig. 2). Thirty-five genes implicated in



**Fig. 2.** (A) Enrichment of CFM-associated genes in the human craniofacial super-enhancers (hCFSEs). The red circles are hCFSE data obtained from NCBI GEO (GSE97752). The labelled red circles show that many CFM-associated genes are enriched in hCFSEs of Carnegie Stages (CS) 13–17; this enrichment is statistically significant. The number in parentheses shows the number of CFM-associated genes enriched in hCFSEs at a specific embryonic time point: for example, at CS 15 (Sample# 13,128), nine previously CFM-associated genes are enriched. (B) Previously reported CFM-associated gene *SHROOM3* is enriched in hCFSEs at CS 13, 14, 15, and 17 (UCSC Genome Browser screenshot of a 100 kb window). (C) Previously reported CFM-associated gene *MYT1* is enriched in hCFSEs at CS 13 and 14 (UCSC Genome Browser screenshot of a 100 kb window).

**Table 1**

Genes with high confidence protein–altering variants and the hCFSEs activity detected in different samples.

| Genes          | hCFSE Samples                                                                                                                                                                                            | hCFSE count |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <i>KIF1B</i>   | CS17_12,341,CS17_12,611,F2_N2108,CS17_12,191,CS20_12,104,CS17_12,331                                                                                                                                     | 6           |
| <i>ZNF423</i>  | CS13_12,877,CS17_12,341,CS13_12,830,CS14_12,408,CS17_12,611,CS15_13,000,CS13_12,383,CS15_13,128,F2_N2108,CS17_12,191,CS15_13,019,CS13_12,690,CS20_12,104,CS17_12,331,CS14_12,709,CS13_12,829,CS14_12,913 | 17          |
| <i>SHROOM3</i> | CS13_12,830,CS13_12,383,CS17_12,191,CS13_12,829                                                                                                                                                          | 4           |
| <i>ZEB1</i>    | CS20_12,104,CS17_12,331                                                                                                                                                                                  | 2           |
| <i>LYST</i>    | CS13_12,877,CS13_12,383                                                                                                                                                                                  | 2           |
| <i>INPPL1</i>  | CS17_12,191,CS20_12,104,CS17_12,331                                                                                                                                                                      | 3           |
| <i>KMT2D</i>   | CS17_12,191,CS20_12,104,CS17_12,331                                                                                                                                                                      | 3           |
| <i>CDH5</i>    | CS15_13,000,CS13_12,383,F2_N2108                                                                                                                                                                         | 3           |
| <i>ITGB1</i>   | CS13_12,877,CS17_12,341,CS13_12,830,CS17_12,611,CS15_13,128,F2_N2108,CS17_12,191,CS15_13,019,CS13_12,690,CS20_12,104,CS17_12,331,CS14_12,709,CS13_12,829                                                 | 13          |
| <i>SIK3</i>    | CS17_12,611,F2_N2108,CS17_12,331                                                                                                                                                                         | 3           |
| <i>NF1</i>     | CS17_12,611,CS13_12,690,CS17_12,331                                                                                                                                                                      | 3           |
| <i>CELSR1</i>  | CS13_12,877,CS13_12,830,CS14_12,408,CS15_13,000,CS13_12,383,CS15_13,128,CS15_13,019,CS13_12,690,CS14_12,709,CS13_12,829,CS14_12,913                                                                      | 11          |
| <i>SHROOM3</i> | CS13_12,830,CS13_12,383,CS17_12,191,CS13_12,829                                                                                                                                                          | 4           |
| <i>USP19</i>   | CS20_12,104                                                                                                                                                                                              | 1           |
| <i>CHD7</i>    | F2_N2108,CS15_13,019,CS13_12,690,CS17_12,331                                                                                                                                                             | 4           |
| <i>EYA1</i>    | CS13_12,877,CS17_12,341,CS17_12,611,CS15_13,128,CS17_12,191,CS13_12,690,CS17_12,331,CS14_12,913                                                                                                          | 8           |
| <i>FOXC1</i>   | CS13–12,877,CS17–12,341,CS13–12,830,CS14–12,408,CS17–12,611,CS15–13,000,CS13–12,383,CS15–13,128,F2–N2108,CS17–12,191,CS15–13,019,CS13–12,690,CS17–12,331,CS14–12,709,CS13–12,829,CS14–12,913             | 16          |
| <i>BPTF</i>    | CS17_12,331                                                                                                                                                                                              | 1           |

CFM pathogenesis were assessed for enrichment at various Carnegie Stages (CS) relevant to human craniofacial development (Fig. 2A). We observed significant enrichment of these CFM associated genes in hCFSEs, particularly at Carnegie Stages 13–17 ( $p < 0.05$ ) (Fig. 2A). For example, *SHROOM3*, a gene strongly implicated in CFM pathogenesis, revealed robust hCFSE activity at CS 13, 14, 15, and 17 within its gene body (Fig. 2B). Similarly, *MYT1* exhibited hCFSE activity upstream of the gene during CS 13 and 14 (Fig. 2C). This finding suggests that the analyzed CFM genes are, perhaps, regulated by hCFSEs during key stages of embryonic craniofacial development.

#### Exome sequencing analyses identify protein–altering variants in genes enriched in hCFSEs

The exome sequencing dataset comprised primarily unilateral CFM cases, with one bilateral presentation. Exome data were generated from buccal swab samples collected from probands and at least one unaffected parent. Maternal DNA was available for all families, and paternal DNA was included for one family. Analysis of exome sequencing data identified high–quality, protein–altering variants in genes within the putative targets of hCFSEs (Table 1). All variants had a minimum read depth (RD) of 10 and genotype quality (GQ) of 20, with minor allele frequency (MAF)  $< 1\%$  in the gnomAD exome population database. To further assess gene function in craniofacial development, we also examined human phenotypes associated with genes affected by copy–number variations (CNVs).

Genes affected by CNVs in individuals with craniofacial disorders, as reported in the DECIPHER database, were also evaluated. Supporting evidence from model organisms indicated that disruption of these genes in mice leads to craniofacial anomalies. In silico prediction tools showed that the identified variants are among the top 10 % most deleterious in the human genome, with CADD scores  $\geq 15$ , a threshold for likely pathogenicity (Table 2).<sup>35</sup> Variants were further classified using the American College of Medical Genetics and Genomics (ACMG) 5–tier system, ranging from benign to pathogenic (Table 2).<sup>40</sup>

#### Damaging variant in *KIF1B* contribute to risk of CFM

Integrative omics analysis identified a protein–altering variant in *KIF1B* (p.Leu842Ser) in an affected individual (Fig. 3A–C). This deleterious variant, located in a gene encoding a kinesin family member protein, was detected in samples from both the affected and unaffected sides of the individual with CFM. Segregation analysis of available family members revealed that the *KIF1B* (p.Leu842Ser) variant was absent in the unaffected mother (Fig. 3C). However, lack of paternal genotype data precluded determination of the variant’s inheritance pattern.

This *KIF1B* variant is extremely rare, identified in only 2 of over 1.4 million exome sequences (MAF = 0.00000137) in the gnomAD database. Conservation analysis showed the affected leucine residue is highly conserved, and the serine substitution has not been observed in homologous sequences; the serine also possesses distinct biochemical properties. The *KIF1B* c.2525T>C (p.Leu842Ser) variant is predicted to be deleterious (SIFT score: 0), probably damaging (PolyPhen2 score: 0.99), and among the top 0.1 % most deleterious mutations in the human genome (CADD score: 29.7). According to ACMG criteria, the variant was classified as likely pathogenic, and ClinGen predicts autosomal dominant (AD) inheritance (<https://search.clinicalgenome.org/kb/gene–validity>).<sup>41</sup>

Constraint metrics for *KIF1B* indicate intolerance to missense and loss–of–function variants (missense Z–score: 5.26; LOEUF score: 0.38), supporting a pathogenic role for the p.Leu842Ser variant. Structural modeling showed that substitution of leucine with serine at position 842 results in a smaller protein, disrupts the  $\alpha$ –helix secondary structure, and leads to loss of molecular interactions. This residue is highly conserved and located within the kinesin functional domain, indicating that the p.Leu842Ser variant likely impairs protein function.

#### Discussion

Due to the low frequency of this birth defect in the population, analyses based on probabilistic models are difficult to conduct, as the available sample sizes are relatively small and underpowered to identify significant risk variants. This explains why there are a few genome–wide association studies.<sup>11,15</sup> Thus, we used a highly reproducible, rigorous, integrative multi–omics approach to identify genetic risk factors underlying the developmental pathogenesis of CFM birth defect. Discoveries from these innovative omics analyses are expected to reduce the knowledge gap in the etiopathogenesis of CFM.

Integrative analysis combining high–throughput epigenomic profiling of genes likely regulated by hCFSEs with exome sequencing in the CFM cohort identified rare variants in genes implicated in craniofacial development. For example, we detected protein–altering variants in the candidate gene *SHROOM3*, which were predicted to be deleterious *in silico* but classified as benign according to ACMG criteria.

Intriguingly, we also identified a protein–altering variant in the *KIF1B* gene, *KIF1B* (KIF1B:p.Leu842Ser). This variant was classified as likely –pathogenic according to the ACMG criteria. This gene is enriched in hCFSEs, and our discovery of this protein–altering variant is the first report of the potential role of *KIF1B* in the developmental pathogenesis of CFM.

The identification of hCFSEs has enabled novel insights into craniofacial development. The hCFSEs are regulatory elements that are active during critical stages of craniofacial morphogenesis. Developmental





mutations in genes which contribute to craniofacial birth defects. This is due to the regulatory mechanisms of these hCFSEs on surrounding genes in craniofacial progenitor cells and at time points specific to craniofacial development.<sup>22</sup> These are part of the gene regulatory machinery which are located between genes or within the intron of a gene. The active state of these hCFSEs has been found to be turned on early during embryonic development, thus sharing similar temporal state with the craniofacial development.<sup>22</sup> Indeed, common variants associated with the most common craniofacial birth defect – orofacial clefts – are significantly enriched in enhancers that are active during early embryogenesis.<sup>22</sup> Hence, it was not surprising to find significant enrichment of known CFM candidate genes in these hCFSEs.

We identified rare protein–altering variants in genes enriched in hCFSEs. These variants have a high CADD score, hence classified to be among the top 1 % most deleterious protein–altering variants in the human genome. Additionally, these deleterious variants were classified using the ACMG guidelines for categorization of variants based on their pathogenicity. These multi–omics analysis led to the discovery of the deleterious *KIF1B* variant. The protein–altering variant disrupts the secondary structure, and this disruptive amino acid change is within the *KIF1B* functional domain. Thus, this integrative analysis led to the discovery of a new CFM candidate gene, *KIF1B*.

*KIF1B* encodes a kinesin family member 1B motor protein involved in transporting mitochondrial and synaptic vesicle precursors. Mutations in *KIF1B* cause hereditary neuropathies affecting peripheral and cranial nerves, including the facial nerve, leading to unilateral or bilateral facial paralysis.<sup>43,44</sup> Genetic manipulation in mice support the role of this gene in developmental pathogenesis of CFM as *Kif1btm1b(KOMP)Wtsi/Kif1btm1b(KOMP)Wtsi* mouse showed an abnormal head shape as well as facial and ear morphology. This finding of a novel CFM candidate gene provides additional evidence for the high yield of multi–omics analysis in the discovery of disease–causing genetic variants contributing to the developmental pathogenesis of birth defects.

## Conclusions

Epigenomic analysis revealed significant enrichment of CFM candidate genes within hCFSEs. Using an integrative omics approach, we identified a likely pathogenic variant in the CFM candidate gene *KIF1B* (c.2525T>C). This strategy demonstrates the potential to uncover novel etiologic variants and improve molecular diagnosis of CFM. Advancing our understanding of CFM molecular pathogenesis through such approaches will inform the development of targeted therapies and may ultimately reduce disease incidence and burden.

## Declaration of generative AI and AI–assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used “Grammarly Pro” program in order to ensure clear, concise, well–structured, grammatically correct and scientifically rigorous writing. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

## Declaration of competing interest

Dr. Veerasathpurush Allareddy serves as an Associate Editor of *Seminars in Orthodontics* and is the Guest Editor of Graber Symposium Edition.

## CRediT authorship contribution statement

**Waheed Awotoye:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Pedro A. Sanchez–Lara:** Writing – review & editing, Writing – original draft, Resources,

Methodology, Investigation, Data curation. **Stephen Yen:** Writing – review & editing, Writing – original draft, Resources, Methodology, Investigation, Conceptualization. **Emily Farrow:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Huojun Cao:** Methodology, Investigation, Formal analysis, Data curation. **Aline Petrin:** Writing – review & editing, Writing – original draft, Visualization, Validation, Data curation. **Lina Moreno Uribe:** Writing – review & editing, Writing – original draft, Investigation. **Veerasathpurush Allareddy:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Shankar Rengasamy Venugopalan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization.

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